

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.
20										821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.
20	U04		MODULAR DBB	K03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
30	U04	VAV	MODULAR DBB	K03AC02	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
40	U04	VAW	MODULAR DBB	M03EC02	0.5	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	1.5	3	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
50										FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
50		VB2					***** BALL VALVES *****			FF5	FLAT BOTTOM NIPPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
60	U04		BALL	A03AC01	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
70	U04		BALL	A03AC02	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0	FW4	Only to be used for fire equipment nozzle connection
80	U04		BALL	A03AC07	6	10	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
90	U04		BALL	A03AC08	6	10	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
100	U04		BALL	B03AC01	0.5	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY
110	U04		BALL	B03AC02	0.5	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
120	U04	VA5	BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UJ3	USE 3IN WITH THERMOWELL CONNECTIONS ONLY
130	U04	VA5	BALL	B03AC04	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	UJ4	FOR 2 INCH THERMOWELL CONNECTIONS ONLY
140	U04	VA5	BALL	B03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
150	U04	VA5	BALL	B03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
160	U04	VA5	BALL	B03JC07	26	48	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
170	U04	VA5	BALL	B03JC08	26	48	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
180							***** RISING STEM BALL VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
180										VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
190	U04	VA5	BALL	R03AC08	10	10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
200							***** BUTTERFLY VALVES *****				
200											
210			BUTTERFLY	T03AC61	10	24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
220			BUTTERFLY	T03JC61	26	48	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0		
230	U04			T06AC61	10	10	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
240							***** GATE VALVES *****				
240											
250											
250	U04		GATE	G03AC48	0.5	1.5	GATE 300# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		
260			GATE	G03AC41	2	8	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0		
270		VA5	GATE	G03AC42	10	24	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE GO	0	0		
280							***** GLOBE VALVES *****				
280											
290											
290	U04		GLOBE	L03AC48	0.75	1.5	GLOBE 300# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
300			GLOBE	L03AC41	2	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
310			GLOBE	L03AC42	8	10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
320							***** CHECK VALVES *****				
320											
330											
330	U04	VC1	CHECK	C03AC50	0.75	1.5	CHECK PISTON 300# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
340	U04		CHECK	C03AC45	2	10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
350	U04		CHECK	C03AC52	12	24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16	0	0		
360	U04		CHECK	C03JC52	26	48	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0		
370							*****PIPES*****				
370											
380											
380			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
390			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
400	U04	UK9	PIPE	5358511	0.5	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
410			PIPE	5358505	3	14	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
420			PIPE	5358504	16	16	PIPE SCH 40 SMLS LTCS A333GR.6 BE	0	0		
430			PIPE	5358598	18	24	PIPE SCH 40 EFW LTCS A671 CC60 CL.22 BE	0	0		
440			PIPE	5358611	26	30	PIPE WT 18MM EFW LTCS A671 CC60 CL.22 BE	0	0		
450			PIPE	5358612	32	32	PIPE WT 20.0MM EFW LTCS A671 CC60 CL.22 BE	0	0		
460	U04	821	PIPE	5358747	24	24	PIPE WT 19.05MM EFW LTCS A671 CC60 CL.22 BE	0	0		
470			PIPE	5358614	34	38	PIPE WT 22MM EFW LTCS A671 CC60 CL.22 BE	0	0		
480			PIPE	5358617	40	42	PIPE WT 25MM EFW LTCS A671 CC60 CL.22 BE	0	0		
490			PIPE	5358619	44	46	PIPE WT 27MM EFW LTCS A671 CC60 CL.22 BE	0	0		
500			PIPE	5358620	48	48	PIPE WT 29MM EFW LTCS A671 CC60 CL.22 BE	0	0		
510		810	PIPE	5358512	0.5	0.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
520							*****FITTINGS*****				
520											
530											

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR WT > 19 MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW: 1.5 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS RATING: 300 RF TEMPERATURE RANGE: -40 / 343 °C	12/16/2024	BY OZAC	U04	Page 1 of 5	300H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
530			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0	
540	U04	UK9	ELBOW	5392815	0.5	0.75	ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6	0	0	
550			ELBOW	5392438	3	14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0	
560			ELBOW	5392384	16	16	ELL 90 DEG LR SCH 40 LTCS A420 WPL6	0	0	
570			ELBOW	5392386	18	24	ELL 90 DEG LR SCH 40 LTCS A420 WPL6-W	0	0	
580			ELBOW	5392530	26	30	ELL 90 DEG LR WT 18.0MM LTCS A420 WPL6-W	0	0	
590			ELBOW	5392535	32	32	ELL 90 DEG LR WT 20.0MM LTCS A420 WPL6-W	0	0	
600			ELBOW	5392531	34	38	ELL 90 DEG LR WT 22.0MM LTCS A420 WPL6-W	0	0	
610			ELBOW	5392532	40	42	ELL 90 DEG LR WT 25.0MM LTCS A420 WPL6-W	0	0	
620			ELBOW	5392533	44	46	ELL 90 DEG LR WT 27.0MM LTCS A420 WPL6-W	0	0	
630			ELBOW	5392534	48	48	ELL 90 DEG LR WT 29.0MM LTCS A420 WPL6-W	0	0	
640			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0	
650	U04	UK9	ELBOW	5392831	0.5	0.75	ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6	0	0	
660			ELBOW	5392444	3	14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0	
670			ELBOW	5392390	16	16	ELL 45 SCH 40 LTCS A420 WPL6	0	0	
680			ELBOW	5392389	18	24	ELL 45 DEG SCH 40 LTCS A420 WPL6 -W	0	0	
690			ELBOW	5392540	26	30	ELL BW 45 DEG LR WT 18.0MM LTCS A420 WPL6 -W	0	0	
700			ELBOW	5392545	32	32	ELL 45 DEG LR WT 20.0MM LTCS A420 WPL6-W	0	0	
710			ELBOW	5392541	34	38	ELL BW 45 DEG LR WT 22.0MM LTCS A420 WPL6 -W	0	0	
720			ELBOW	5392542	40	42	ELL BW 45 DEG LR WT 25.0MM LTCS A420 WPL6 -W	0	0	
730			ELBOW	5392543	44	46	ELL BW 45 DEG LR WT 27.0MM LTCS A420 WPL6 -W	0	0	
740			ELBOW	5392544	48	48	ELL 45 DEG LR WT 29.0MM LTCS A420 WPL6-W	0	0	
750										
750			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0	
760	U04	UK9	TEE	5392879	0.5	0.75	TEE SCH XXS LTCS A420 WPL6	0	0	
770			TEE	5392447	3	14	TEE BW SCH STD LTCS A420 WPL6	0	0	
780			TEE	5392393	16	16	TEE SCH 40 LTCS A420 WPL6	0	0	
790			TEE	5392400	18	24	TEE SCH 40 LTCS A420 WPL6-W	0	0	
800			TEE	5392408	26	30	TEE WT 18.0MM LTCS A420 WPL6-W	0	0	
810			TEE	5392415	32	32	TEE BW WT 20.0MM LTCS A420 WPL6-W	0	0	
820			TEE	5392409	34	38	TEE WT 22.0MM LTCS A420 WPL6-W	0	0	
830			TEE	5392410	40	42	TEE WT 25.0MM LTCS A420 WPL6-W	0	0	
840			TEE	5392411	44	46	TEE WT 27.0MM LTCS A420 WPL6-W	0	0	
850			TEE	5392414	48	48	TEE BW WT 29.0MM LTCS A420 WPL6-W	0	0	
860			TEE RED	5392661	0.5	1.5	TEE RED BW SCH XS LTCS A420 WPL6	0.75	2	
870			TEE RED	5392454	1.5	1.5	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3	
880			TEE RED	5392454	2	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	4	
890			TEE RED	5392445	3	6	TEE RED BW SCH STD LTCS A420 WPL6	4	14	
900			TEE RED	5392445	8	12	TEE RED BW SCH STD LTCS A420 WPL6	10	14	
910			TEE RED	5392293	6	14	TEE RED SCH 40/STD LTCS A420 WPL6	16	16	
920			TEE RED	5392401	8	14	TEE RED SCH 40/STD LTCS A420 WPL6-W	18	24	
930	U04		TEE RED	5392474	16	20	TEE RED SCH 40 LTCS A420 WPL6-W	20	24	
940			TEE RED	5393013	16	24	TEE RED WT 18MM/SCH 40 LTCS A420 WPL6-W	26	30	
950			TEE RED	5392350	18	24	TEE RED WT 22.0MM/SCH 40 LTCS A420 WPL6-W	34	38	
960			TEE RED	5392201	26	28	TEE RED WT 18.0MM LTCS A420 WPL6-W	28	30	
970			TEE RED	5392203	26	30	TEE RED WT 20.0MM/18.0MM LTCS A420 WPL6-W	32	32	
980			TEE RED	5392204	32	32	TEE RED WT 22.0MM/ 20.0MM LTCS A420 WPL6-W	34	38	
990			TEE RED	5392323	34	36	TEE RED WT 22MM LTCS A420 WPL6-W	36	38	
1000			TEE RED	5392326	34	38	TEE RED WT 25.0MM/22.0MM LTCS A420 WPL6-W	40	42	
1010			TEE RED	5392324	40	40	TEE RED WT 25.0MM LTCS A420 WPL6-W	42	42	
1020			TEE RED	5392207	34	38	TEE RED WT 27.0MM/22.0MM LTCS A420 WPL6-W	44	46	
1030			TEE RED	5392327	40	42	TEE RED WT 27.0MM/25MM LTCS A420 WPL6-W	44	46	
1040			TEE RED	5392325	44	44	TEE RED WT 27.0MM LTCS A420 WPL6-W	46	46	
1050			TEE RED	5392697	40	42	TEE RED WT 29.0MM/25.0MM LTCS A420 WPL6-W	48	48	
1060			TEE RED	5392205	44	46	TEE RED WT 29.0MM/ 27.0MM LTCS A420 WPL6-W	48	48	
1070										
1070			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0	
1080			CAP	5392448	3	14	CAP BW SCH STD LTCS A420 WPL6	0	0	
1090			CAP	5392394	16	24	CAP SCH 40 LTCS STL A420 WPL6	0	0	
1100			CAP	5392616	26	30	CAP WT 18.0MM LTCS A420 WPL6	0	0	
1110			CAP	5392625	32	32	CAP WT 20.0MM LTCS A420 WPL6	0	0	
1120			CAP	5392617	34	38	CAP WT 22.0MM LTCS A420 WPL6	0	0	
1130			CAP	5392618	40	42	CAP WT 25.0MM LTCS A420 WPL6	0	0	
1140			CAP	5392619	44	46	CAP WT 27.0MM LTCS A420 WPL6	0	0	
1150			CAP	5392624	48	48	CAP WT 29.0MM LTCS A420 WPL6	0	0	
1160										
1160	U04		RED CONC	5392458	0.5	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	0.75	2	
1170			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	4	
1180			RED CONC	5392449	3	12	REDUCER CONC SCH STD LTCS A420 WPL6	4	14	
1190			RED CONC	5392300	8	14	REDUCER CONC SCH 40/STD LTCS A420 WPL6	16	16	
1200			RED CONC	5392301	10	14	REDUCER CONC SCH 40/STD LTCS A420 WPL6-W	18	20	
1210			RED CONC	5392395	16	20	REDUCER CONC SCH 40 LTCS A420 WPL6-W	18	24	
1220			RED CONC	5392359	20	24	REDUCER CONC 18.0MM/SCH 40 LTCS A420 WPL6-W	26	30	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR WT > 19 MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW: 1.5 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS RATING: 300 RF TEMPERATURE RANGE: -40 / 343 °C	12/16/2024	BY OZAC	U04	Page 2 of 5	300H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1230			RED CONC	5392971	24	24	REDUCER CONC WT 20.0MM/SCH 40 LTCS A420 WPL6-W	32	32	
1240			RED CONC	5392360	24	24	REDUCER CONC 22.0MM/SCH 40 LTCS A420 WPL6-W	34	36	
1250			RED CONC	5392345	26	30	REDUCER CONC WT 20.0MM/18.0MM LTCS A420 WPL6-W	32	32	
1260			RED CONC	5392355	26	30	REDUCER CONC 22.0MM/18.0MM LTCS A420 WPL6-W	34	38	
1270			RED CONC	5392356	30	30	REDUCER CONC 25.0MM/18.0MM LTCS A420 WPL6-W	40	42	
1280			RED CONC	5392357	34	38	REDUCER CONC 25.0MM/22.0MM LTCS A420 WPL6-W	40	42	
1290			RED CONC	5392347	36	38	REDUCER CONC WT 27.0MM/22.0MM LTCS A420 WPL6-W	44	46	
1300			RED CONC	5392358	40	42	REDUCER CONC WT 27.0MM/25.0MM LTCS A420 WPL6-W	44	46	
1310			RED CONC	5392997	40	42	REDUCER CONC WT 29.0MM/25.0MM LTCS A420 WPL6-W	48	48	
1320			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2	
1330			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	4	
1340			RED ECC	5392450	3	12	REDUCER ECC SCH STD LTCS A420 WPL6	4	14	
1350			RED ECC	5392302	8	14	REDUCER ECC SCH 40/STD LTCS A420 WPL6	16	16	
1360			RED ECC	5392303	10	14	REDUCER ECC SCH 40/STD LTCS A420 WPL6-W	18	20	
1370			RED ECC	5392396	16	20	REDUCER ECC SCH 40 LTCS A420 WPL6-W	18	24	
1380			RED ECC	5392517	20	24	REDUCER ECC 18.0MM/SCH 40 LTCS A420 WPL6-W	26	30	
1390	U04	821	RED ECC	5392277	16	16	REDUCER ECC WT 19.05 MM/SCH 40 LTCS A420 WPL6-W	24	24	
1400			RED ECC	5392972	24	24	REDUCER ECC WT 20.0MM/SCH 40 LTCS A420 WPL6-W	32	32	
1410			RED ECC	5392518	24	24	REDUCER ECC 22.0MM/SCH 40 LTCS A420 WPL6-W	34	38	
1420			RED ECC	5392868	26	30	REDUCER ECC WT 20.0MM/18.0MM LTCS A420 WPL6-W	32	32	
1430			RED ECC	5392525	26	30	REDUCER ECC 22.0MM/18.0MM LTCS A420 WPL6-W	34	38	
1440			RED ECC	5392526	30	30	REDUCER ECC 25.0MM/18.0MM LTCS A420 WPL6-W	40	42	
1450			RED ECC	5392527	34	38	REDUCER ECC 25.0MM/22.0MM LTCS A420 WPL6-W	40	42	
1460			RED ECC	5392565	36	38	REDUCER ECC WT 27.0MM/22.0MM LTCS A420 WPL6-W	44	46	
1470			RED ECC	5392528	40	42	REDUCER ECC WT 27.0MM/25.0MM LTCS A420 WPL6-W	44	46	
1480			RED ECC	5392998	40	42	REDUCER ECC WT 29.0MM/25.0MM LTCS A420 WPL6-W	48	48	
1490										
1490	U04		OLETS	5504020	0.75	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	2	2	
1500	U04		OLETS	5504021	0.75	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	14	
1510	U04		OLETS	5504022	0.75	2	NIPOFLANGE 300# RF SCH 40/XS LTCS A350 LF2 CL.1	16	24	
1520	U04		OLETS	5504104	0.75	2	NIPOFLANGE 300# RF WT 18.0MM/XS LTCS A350 LF2 CL.1	26	30	
1530	U04		OLETS	5504105	0.75	2	NIPOFLANGE 300# RF 20.0MM/XS LTCS A350 LF2 CL.1	32	32	
1540	U04		OLETS	5504106	0.75	2	NIPOFLANGE 300# RF WT 22MM/XS LTCS A350 LF2 CL.1	34	38	
1550	U04		OLETS	5504107	0.75	2	NIPOFLANGE 300# RF WT 25MM/XS LTCS A350 LF2 CL.1	40	42	
1560	U04		OLETS	5504108	0.75	2	NIPOFLANGE 300# RF WT 27MM/XS LTCS A350 LF2 CL.1	44	46	
1570	U04		OLETS	5504109	0.75	2	NIPOFLANGE 300# RF WT 29MM/XS LTCS A350 LF2 CL.1	48	48	
1580	U04	FF5	OLETS	5504501	0.75	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A350 LF2 CL.1	0	0	
1590	U04	UJ2	OLETS	5504025	1.5	1.5	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	14	
1600	U04	UJ2	OLETS	5504018	1.5	1.5	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2 CL.1	16	24	
1610	U04	UJ2	OLETS	5504114	1.5	1.5	NIPOFLANGE 300# RF WT 18.0MM/160 LTCS A350 LF2 CL.1	26	30	
1620	U04	UJ2	OLETS	5504115	1.5	1.5	NIPOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2 CL.1	32	32	
1630	U04	UJ2	OLETS	5504116	1.5	1.5	NIPOFLANGE 300# RF WT 22.0MM/160 LTCS A350 LF2 CL.1	34	38	
1640	U04	UJ2	OLETS	5504117	1.5	1.5	NIPOFLANGE 300# RF WT 25MM/160 LTCS A350 LF2 CL.1	40	42	
1650	U04	UJ2	OLETS	5504118	1.5	1.5	NIPOFLANGE 300# RF WT 27MM/160 LTCS A350 LF2 CL.1	44	46	
1660	U04	UJ2	OLETS	5504119	1.5	1.5	NIPOFLANGE 300# RF WT 29.0MM/160 LTCS A350 LF2 CL.1	48	48	
1670	U04	UJ3	OLETS	5504059	3	3	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2 CL.1	18	20	
1680	U04	UJ4	OLETS	5504110	2	2	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2 CL.1	16	16	
1690	U04	UJ4	OLETS	5504110	2	2	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2 CL.1	20	20	
1700	U04	UJ3	OLETS	5504325	3	3	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	8	8	
1710	U04	UJ4	OLETS	5504326	2	2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	8	10	
1720	U04	UJ4	OLETS	5504326	2	2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	14	14	
1730	U04	UJ3	OLETS	5505234	3	3	NIPOFLANGE 300# RF WT 18.0MM/160 LTCS A350 LF2 CL.1	30	30	
1740	U04	UJ3	OLETS	5504515	3	3	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2 CL.1	0	0	
1750										
1750	U04		OLETS	5505335	0.75	1	WELDOLET SCH XS LTCS A350 LF2 CL.1	2	2	
1760	U04		OLETS	5505179	0.75	1	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	14	
1770	U04		OLETS	5505179	1.5	1.5	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	4	14	
1780	U04		OLETS	5505179	2	2	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	6	14	
1790	U04		OLETS	5505574	0.75	2	WELDOLET SCH 40/XS LTCS A350 LF2 CL.1	16	24	
1800	U04		OLETS	5505579	0.75	2	WELDOLET WT 18MM/ SCH XS LTCS A350 LF2 CL.1	26	30	
1810	U04		OLETS	5505582	0.75	2	WELDOLET WT 20.0MM/XS LTCS A350 LF2 CL.1	32	32	
1820	U04		OLETS	5505585	0.75	2	WELDOLET WT 22MM/XS LTCS A350 LF2 CL.1	34	38	
1830	U04		OLETS	5505587	0.75	2	WELDOLET WT 25MM/XS LTCS A350 LF2 CL.1	40	42	
1840	U04		OLETS	5505589	0.75	2	WELDOLET WT 27MM/XS LTCS A350 LF2 CL.1	44	46	
1850	U04		OLETS	5505597	0.75	2	WELDOLET WT 29.0MM/XS LTCS A350 LF2 CL.1	48	48	
1860	U04		OLETS	5505164	3	4	WELDOLET SCH STD LTCS A350 LF2 CL.1	6	14	
1870	U04		OLETS	5505554	3	8	WELDOLET SCH 40/STD LTCS A350 LF2 CL.1	16	24	
1880	U04		OLETS	5505580	3	12	WELDOLET 18.0MM/ STD LTCS A350 LF2 CL.1	26	30	
1890	U04		OLETS	5505583	3	12	WELDOLET WT 20.0MM/STD LTCS A350 LF2 CL.1	32	32	
1900	U04		OLETS	5505586	3	12	WELDOLET SCH 22.0MM/STD LTCS A350 LF2 CL.1	34	38	
1910	U04		OLETS	5505588	3	12	WELDOLET WT 25.0MM/STD LTCS A350 LF2 CL.1	40	42	
1920	U04		OLETS	5505590	3	12	WELDOLET SCH 27.0MM/STD LTCS A350 LF2 CL.1	44	46	
1930	U04		OLETS	5505598	3	12	WELDOLET WT 29.0MM/ STD LTCS A350 LF2 CL.1	48	48	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR WT > 19 MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW: 1.5 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS RATING: 300 RF TEMPERATURE RANGE: -40 / 343 °C	12/16/2024	BY OZAC	U04	Page 3 of 5	300H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1940	U04		OLETS	5504374	0.75	0.75	WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1	0	0	
1950	U04		OLETS	5504374	1.5	2	WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1	0	0	
1960										
1960	U04		ELBOFLG	5516625	1.5	1.5	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	14	
1970	U04		ELBOFLG	5516618	1.5	1.5	ELBOFLANGE 300# RF SCH 40/160 LTCS A350 LF2 CL.1	16	24	
1980	U04		ELBOFLG	5516684	1.5	1.5	ELBOFLANGE 300# RF WT 18.0MM/160 LTCS A350 LF2 CL.1	26	30	
1990	U04		ELBOFLG	5516685	1.5	1.5	ELBOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2 CL.1	32	32	
2000	U04		ELBOFLG	5516686	1.5	1.5	ELBOFLANGE 300# RF WT 22.0MM/160 LTCS A350 LF2 CL.1	34	38	
2010	U04		ELBOFLG	5516687	1.5	1.5	ELBOFLANGE 300# RF WT 25.0MM/160 LTCS A350 LF2 CL.1	40	42	
2020	U04		ELBOFLG	5516688	1.5	1.5	ELBOFLANGE 300# RF WT 27.0MM/160 LTCS A350 LF2 CL.1	44	46	
2030	U04		ELBOFLG	5516689	1.5	1.5	ELBOFLANGE 300# RF WT 29.0MM/160 LTCS A350 LF2 CL.1	48	48	
2040										
2040	U04	821	OLETS	5509731	3	3	SWEEPOLET 19.05 MM/STD LTCS A350 LF2 CL.1	24	24	
2050										
2050							*****PLUGS*****			
2060										
2060	U04	UDC	FITTING SCRD	5491303	0.5	0.5	PLUG ROUND HEAD SCRD LTCS A350 LF2 CL.1	0	0	
2070										
2070		FS1					*****FLANGES*****			
2080										
2080	U04		FLANGE	5556476	0.5	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2090	U04		FLANGE	5556473	3	14	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
2100	U04		FLANGE	5556472	16	20	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 40 BORE	0	0	
2110	U04		FLANGE	5556472	24	24	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 40 BORE	0	0	
2120	U04		FLANGE	5556479	26	30	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 18.0MM B16.47 SERIES A	0	0	
2130	U04		FLANGE	5556492	32	32	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 20.0MM B16.47 SERIES A	0	0	
2140	U04		FLANGE	5556483	34	38	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 22.0MM B16.47 SERIES A	0	0	
2150	U04		FLANGE	5556484	40	42	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 25.0MM B16.47 SERIES A	0	0	
2160	U04		FLANGE	5556485	44	46	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 27.0MM B16.47 SERIES A	0	0	
2170	U04		FLANGE	5556498	48	48	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 29.0MM B16.47 SERIES A	0	0	
2180	U04		FLANGE	5557393	2	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2190	U04		FLANGE	5556967	16	16	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 40	0	0	
2200	U04		FLANGE	5557150	16	16	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 40	0	0	
2210	U04		FLANGE	5557141	0.75	0.75	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2220	U04		BLIND	5615576	0.5	24	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
2230	U04			5615598	10	10	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0	
2240	U04		BLIND	5615579	26	48	FLG BLIND 300# RF LTCS A350 LF2 CL.1 B16.47 SERIES A	0	0	
2250	U04		FLANGE	5557036	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2260	U04		FLANGE	5557033	3	14	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
2270	U04	FA8	TEE RED	5599703	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5	
2280										
2280	U04		FLG ADAPTER	5851289	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 6 MM TUBE	0	0	
2290			FLANGE	5851203	0.5	3	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
2300										
2300							***** ORIFICE FLANGES *****			
2310										
2310	U04		ORIFICE FLG	5623947	2	2	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
2320	U04		ORIFICE FLG	5623944	3	14	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
2330	U04		ORIFICE FLG	5623943	16	24	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 40	0	0	
2340	U04		ORIFICE FLG	5624831	26	28	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 18.0MM WT ASME B16.47 SERIES A	0	0	
2350	U04		ORIFICE FLG	5624806	30	30	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 18.0MM WT ASME B16.47 SERIES A	0	0	
2360	U04		ORIFICE FLG	5624832	32	32	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 20.0MM WT ASME B16.47 SERIES A	0	0	
2370	U04		ORIFICE FLG	5624808	34	34	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 22.0MM WT ASME B16.47 SERIES A	0	0	
2380	U04		ORIFICE FLG	5624809	36	36	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 22.0MM WT ASME B16.47 SERIES A	0	0	
2390	U04		ORIFICE FLG	5624810	38	38	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 22.0MM WT ASME B16.47 SERIES A	0	0	
2400	U04		ORIFICE FLG	5624811	40	40	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 25.0MM WT ASME B16.47 SERIES A	0	0	
2410	U04		ORIFICE FLG	5624833	44	44	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 27.0MM WT ASME B16.47 SERIES A	0	0	
2420	U04		ORIFICE FLG	5624813	46	46	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 27.0MM WT ASME B16.47 SERIES A	0	0	
2430	U04		ORIFICE FLG	5624834	48	48	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 29.0MM WT ASME B16.47 SERIES A	0	0	
2440										
2440							***** PIPE NIPPLES *****			
2450										
2450	U04	FW4	NIPPLE	5541187	2	2	NIPPLE SCH XS/160 LTCS A333-6 BOE/TOE 3" LG	0	0	
2460	U04	FW4	NIPPLE	5541184	1	1.5	NIPPLE SCH XS/160 LTCS A333-6 POE/TOE 3" LG	0	0	
2470										
2470		FS1					***** GASKETS *****			
2480										
2480			GASKET	5673053	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2490			GASKET	5674032	26	48	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2500		SB6	GASKET	5673103	0.5	16	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2510	U04		GASKET	5672251	2	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2520	U04	GA6	GASKET	5673017	2	3	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
2530	U04		GASKET	5673117	0.75	16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	

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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2540										
2540		B1					***** BOLTING *****			
2550										
2550		VA5	BOLTING	5675669	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M NUTS	0	0	
2560										
2560		SB6	BOLTING	5675675	12	48	JACK SCREW HEX HEAD A320 GR L7M	0	0	
2570										
2570							**** BLINDS & SPACERS ****			
2580										
2580			SPEC BLIND	5749581	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
2590										
2590			PADDLE	5750100	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
2600			PADDLE	5750028	26	48	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
2610			PADDLE	5750200	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
2620			PADDLE	5750198	26	48	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
2630										

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